

Machine Learning in Quantum Many-Body Physics

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1 Abstract

The complexity of the many-body problem in quantum physics stems from the exponential complexity of the many-body wave function. In this work, I demonstrate that systematic machine learning of the wave function can simplify this complexity into a simpler computational format for certain significant cases of physical relevance. I introduce a variational representation of quantum states based on artificial neural networks with a variable number of hidden neurons. I then describe the applications of ML methods in quantum many-body systems, including tensor networks and the restricted Boltzmann machines. In each section, I describe the basic content related to the definitions and specific procedures for solving the ground state energy.

2 Introduction

Quantum many-body physics refers to a mathematical framework for studying the collective behaviour of large numbers of interacting particles. It can produce an astonishing variety of phases of matter by emerging from apparently simple interactions, including conventional metals and magnetically ordered states, as well as unexpected states such as high-temperature superconductivity, exotic metals, and spin liquids. Quantum many-body systems pose complex and esoteric challenges.

In recent years, machine learning has become increasingly prominent and is widely used in various fields. The application of machine learning in condensed matter physics is particularly promising because it can lead to discoveries and practical solutions to complex physical problems. Machine learning offers a comprehensive set of tools to address numerous challenges in the field, including distinguishing between classical and quantum phases of matter, aiding in the classification and creation of materials, improving existing numerical methods, and potentially laying the foundation for new approaches to quantum many-body physics. The convergence of machine learning and condensed matter physics heralds a new era of innovative exploration and potential breakthroughs that could transform our understanding of key physics concepts.

2.1 What is machine learning?

2.1.1 Definition of machine learning

Machine learning is the use of statistical algorithms by machines to learn from large amounts of historical data. These empirical models then guide the machine's decision-making process in a business environment. This interdisciplinary field explores computer modelling of human learning behaviour, enabling computers to acquire new knowledge, master skills, and improve performance by restructuring existing knowledge structures. At its core, machine learning is

the application of algorithms to examine data, extract knowledge from it, and then use that knowledge to make decisions or predict new data. In another word, the computer uses the acquired data to create a model and then uses that model to make predictions.

2.1.2 Basic concepts in machine learning

Supervised learning: Supervised learning revolves around deducing a function from a labeled training dataset to forecast a target output vector y , given an input vector x . These vectors can encompass either discrete or continuous values. The primary branches of supervised learning are classification, which employs algorithms to precisely classify test data, and regression, which deals with continuous output vectors. In classification, the algorithm discerns and allocates labels to entities within the dataset, whereas regression endeavors to predict a continuous target variable. Linear regression and decision trees stand out as notable methods extensively employed in supervised learning tasks.

A decision tree acts as a classifier, known as a recursive partition of instance space. Geometrically, it can be conceptualized as a collection of hyperplanes, each perpendicular to one of the axes. Decision trees can handle large amounts of data, missing values, and redundant attributes. They have a good ability to generalize and are robust to noise. They also provide high performance for relatively low computational cost.

Linear regression establishes relationships between variables by modeling the link between a continuous dependent variable y and one or more explanatory variables X through a linear function. In the context of regression analysis, the goal is to predict a continuous target variable. In contrast, classification focuses on predicting labels from a finite set. In multiple regression, where a linear combination of input variables is considered, the model takes the form of $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \epsilon$. Similar to decision trees, linear regression is a supervised learning algorithm where the model is trained on labelled data (training data) and then used to predict labels for unlabelled data (test data).

Neural networks: A neural network is a computational model comprised of numerous interconnected nodes, or neurons, each representing a specific activation function. The three main types of neural networks are Feed-Forward Neural Networks (FFNN), Convolutional Neural Networks (CNN), and Recurrent Neural Networks (RNN). FFNNs, trained using the back-propagation learning algorithm, are the most widely used.

The FFNN operates in two distinct modes: training and prediction. Each phase necessitates its own set of data - the training set and the set designated for prediction (the test set). During prediction, information flows through the network from inputs to outputs, handling one example at a time. This process generates estimates of output values based on input values, with the ensuing error serving as a metric for the prediction accuracy of the trained network.

CNNs utilize a mathematical operation known as ‘convolution’ to process data with a grid-like topology. Parameter estimation in CNNs uses the principle of maximum likelihood, where model parameters are selected by assigning a high probability to the observed data. It is more convenient to define a loss function, expressed as the negative logarithm of the likelihood

$$E(\theta) = -\ln(p(y|\theta)) = \sum_{n=1}^M y_n \ln(P_\theta(y|\delta_n)) + (1 - y_n) \ln[1 - P_\theta(y|\delta_n)] \quad (1)$$

Gradient descent techniques are then employed to minimize $E(\boldsymbol{\theta})$ during model training. It is possible to evaluate the gradients analytically by using the chain rule, and it is more practical and less error-prone to use automatic differentiation (AD). AD is a set of techniques for numerically evaluating the derivatives of a function specified by a computer program.

Recurrent Neural Networks (RNNs) are a specialized type of neural network architecture crafted to excel in detecting patterns within sequential data. Unlike conventional networks, where outcomes are exclusively determined by a fixed set of weights, RNNs draw insights not only from these weights but also from the complete temporal sequence of states. This unique attribute equips RNNs to capture and analyze dependencies among different points in the input sequence, making them exceptionally well-suited for tasks involving time-series data or any application where understanding the order of events is critical.

Unsupervised learning: Unsupervised learning delves into various pattern recognition challenges by harnessing unlabeled training samples. It proves particularly adept in situations characterized by unstructured or ungrouped data. Principal Component Analysis (PCA) stands as a pivotal algorithm in this domain, excelling in extracting essential features and reducing dimensionality while retaining crucial information. PCA not only aids in visualizing data but also improves the ability to identify underlying patterns and structures within the dataset.

By identifying the most significant features or components from the input data, PCA aids in simplifying complex datasets while maintaining their integrity. This dimensionality reduction not only streamlines computational processes but also mitigates the curse of dimensionality and improve the efficiency and effectiveness of subsequent analyses. Moreover, PCA's ability to uncover latent relationships between variables further enhances its utility in various applications, ranging from image processing to financial modeling.

2.2 What is quantum many-body physics?

In the field of quantum many-body physics, scientists embark on a quest to decode the intricate phenomena of countless quantum particles as they interact. However, there is a formidable challenge which is to devise the sensitive numerical tools which can grapple with the intense correlations in these systems. Here, the Hilbert space expands exponentially with system size, presenting a daunting obstacle in capturing the essence of the system's state.

The wave function is the most important part in quantum mechanics, denoted as $\psi(r)$. This mathematical entity provides a probabilistic insight into the possible configurations and states that a system can assume. The dynamics of the wave function is governed by the Schrödinger equation, a fundamental principle of quantum mechanics. The time-dependent Schrödinger equation details how the wave function evolves with respect to the Hamiltonian operator, which represents the total energy of the system. Mathematically, the time-dependent Schrödinger equation is expressed as:

$$i\frac{\partial\psi(q,t)}{\partial t} = \hat{H}\psi \quad (2)$$

Representations of wave functions of various physical many-body systems can be succinctly realised with an information content considerably smaller than the maximum capacity of the corresponding Hilbert space. Many-body wave functions are mappings from an N-dimensional set S to an array of complex numbers. Although the number of complex numbers grows

exponentially, they provide a comprehensive description of the amplitude and phase of the quantum state. Typically, a many-body wave function can be formulated as the sum of a set of orthogonal bases and can be completely described by a single function. This function takes a basis as input and the corresponding coefficients of the wave function as output.

Within quantum mechanics' mathematical framework, the concept of Hilbert space holds immense importance. These spaces, as inner product spaces achieving completeness regarding the norm defined by the inner product, are fundamental tools. They deepen our understanding of concepts like Fourier expansion and specific linear transformations, such as the Fourier transform. Also, play a crucial role in quantum mechanics. In the context of quantum mechanics, Hilbert space is essential for handling the infinite-dimensional aspect of quantum states. It enables the representation of a wide range of potential states and configurations. This mathematical framework provides the foundation necessary for describing the probabilistic behaviour and evolution of quantum systems, offering a rigorous basis for the theory's predictions and applications.

In quantum many-body physics, harnessing Hilbert space empowers the application of potent mathematical tools to characterize and scrutinize the dynamics of extensive collections of quantum particles. Crafting sophisticated models and computational techniques becomes necessary for unlocking insights into the intricate phenomena manifested by quantum many-body systems.

2.3 The Many-body Quantum Wave Function

A complete description of a quantum system is given in quantum mechanics by its wave function, alternatively known as its state. The quantum state is simply a vector in finite-dimensional Hilbert spaces.

Physicists utilize the 'ket' notation, where a vector ψ is represented as $|\psi\rangle \in \mathcal{H}$. The Hilbert space $\mathcal{H}$ possesses an inner product denoted by $\langle\phi|\psi\rangle$, which maps a pair of vectors in $\mathcal{H}$ to a scalar. The 'bra' notation, $\langle\phi|$, is employed for the 'dual vector', which formally acts as a linear mapping from vectors to scalars, defined as $|\psi\rangle \mapsto \langle\phi|\psi\rangle$. Conceptually, a 'ket' can be envisioned as a column vector, while a 'bra' resembles a row vector.

One can represent a general single-particle state $|\psi\rangle \in \mathcal{H}_1$, where $\dim(\mathcal{H}_1) = M$, as a linear combination of some orthonormal basis vectors:

$$|\psi\rangle = \sum_{d=1}^M \nu_d |\hat{\psi}_d\rangle \quad (3)$$

where $\mathbf{v} \in \mathbb{C}^M$ is the vector of coefficients compatible with the basis $\{|\hat{\psi}_d\rangle\}_{d=1}^M$ of $\mathcal{H}_1$, each entry of which can be calculated by the projection: $\nu_d = \langle\hat{\psi}_d|\psi\rangle$.

In addition, extend to the many-body case of N particles, each corresponding to a local Hilbert space $\mathcal{H}_j$ for $j \in [N]$ such that, for all j , $\dim(\mathcal{H}_j) = M$. Denoting an orthonormal basis of the local Hilbert space by $\{|\hat{\psi}_d\rangle\}_{d=1}^M$, the many-body wave function $|\psi\rangle \in \mathcal{H} = \otimes_{j=1}^N \mathcal{H}_j$ can be written as

$$|\psi\rangle = \sum_{d_1, \dots, d_N=1}^M \mathcal{A}_{d_1, \dots, d_N} |\hat{\psi}_{d_1, \dots, d_N}\rangle \quad (4)$$

where $|\hat{\psi}_{d_1, \dots, d_N}\rangle := |\hat{\psi}_{d_1}\rangle \otimes \dots \otimes |\hat{\psi}_{d_N}\rangle$ is a basis vector of the M^N -dimensional many-body Hilbert space $\mathcal{H}$, and the coefficient tensor $\mathcal{A}$ is the generalization of the above coefficients vector to the many-body case.

2.4 Main machine learning algorithm used to solve quantum many-body problem

Neural Network Quantum States (NQS or NNQS) constitute a versatile category of variational quantum states parameterized by artificial neural networks. Unlike being confined to ground state problems, NQS can expand to tackle the time-dependent Schrödinger equation. In this setup, the neural network is governed by a distinct set of parameters, yielding a complex-valued output that signifies the amplitude of the wave function. Leveraging this variational framework alongside specific stochastic learning techniques, NQS aims to approximate the desired quantum state. Practically, NQS finds utility in learning the ground state wave function, offering an approximate portrayal aligned with a given Hamiltonian. The learning journey entails determining optimal neural network weights to minimize variational energy, a crucial endeavour given the exponential complexity entailed in computing expectation values for general artificial neural networks. This iterative learning process involves updating neural network parameters and generating a fresh set of samples, similar to unsupervised learning methodologies.

In order to demonstrate the accuracy of Neural Quantum States (NQS) in characterising complex many-body quantum states, our initial focus was on determining the most effective neural network representation for the complicated ground state of a given Hamiltonian H . We then used NQS to characterise the many-body many-body quantum states. Notably, the images depicting the error of NQS ground-state energy relative to the exact value

$$\epsilon_{\text{rel}} = \frac{E_{\text{NQS}}(\alpha) - E_{\text{exact}}}{|E_{\text{exact}}|} \quad (5)$$

reveal a remarkable outcome—the NQS attains a controllable and arbitrary accuracy, aligning with the power-law behaviour of α where α is the density.

2.5 Other algorithms used to solve quantum many-body system

Tensor networks serve as a powerful tool for the entanglement structure in quantum many-body states, encapsulating their local interactions. At the core of numerical algorithms revolving around tensor networks (TNs) lies the Singular Value Decomposition (SVD), intimately connected to the Schmidt decomposition of quantum states. Augmenting this notion is the concept of canonical form, where all bond indices align with orthonormal Hilbert spaces, providing a mathematical underpinning for the efficient representation of quantum states within tensor network frameworks. Traditionally, TN proves particularly effective for states with limited entanglements, such as the ground states of local and gapped Hamiltonians.

A type of generative stochastic artificial neural network called a Restricted Boltzmann Machine (RBM) can learn information about the probability distribution within its input set. RBMs can approximate any desired probability distribution with a high degree of accuracy. This property suggests the potential use of RBMs as approximate representations of many-body wave functions, often referred as neural network quantum states. The advantages and disadvantages of RBMs over Tensor Networks (TNs) will depend on the specific structure of the wavefunction.

3 Application of Tensor Networks in quantum many-body physics

In recent decades, tensor networks have been considered as a novel class of methodologies for investigating strongly correlated systems within low-dimensional lattices. Tensor Network is not only the numerical algorithms, it has also provided a language to comprehensively grasp such systems analytically.

3.1 Definition of Tensor

Tensors lead to various definitions and have been applied across physics, mathematics, and computer science. From a practical standpoint, tensors can be conceptualized as multidimensional arrays. Visually, we can depict them as a shape with lines emanating from it and each line represents an index. For example, we can represent a scalar, a vector, and a matrix as



Figure 1: A scalar, a vector, and a matrix

The number of indices is referred to as the rank of a tensor. By connecting the lines of different tensors, we can represent a summation (or contraction) over that index.

Mathematically, tensor is defined as a series of numbers labeled by N indexes, with N called the order of the tensor. The tensor plays an important role in the wave function $\psi(r)$, in the sense that the whole wave function can be reconstructed from this fundamental piece. A tensor could be represented as a potentially multidimensional array. For example, a tensor can be written as

$$\hat{\mathbf{e}}_i = \sum_{j=1}^n \mathbf{e}_j R_i^j = \mathbf{e}_j R_i^j. \quad (6)$$

This example shows the new basis vectors $\hat{\mathbf{e}}_x$ are expressed in terms of the old basis vectors $\mathbf{e}_j R_i^j$ are the entries of the change of basis matrix. And in the rightmost expression the summation sign is the Einstein summation convention, which is widely used throughout mathematics and physics.

3.2 Definition of Tensor Network

Tensor Network is defined as the contraction of many tensors with a general form

$$\mathcal{T}_{\{s\}} = \sum_{\{a\}} \prod_n T_{s_1^n s_2^n \dots, a_1^n a_2^n \dots}^{[n]}, \quad (7)$$

The indices $\{a\}$ are geometrical indices, each shared by two tensors and subject to contraction. On the other hand, the indices $\{s\}$ are open bonds, with each belonging exclusively to one tensor. Upon contracting all geometrical indices, the Tensor Network represents an N -th order tensor, with N being the total number of open indices $\{s\}$. $\{T^{[n]}\}$ denotes specific tensors. To

delve deeper, Tensor Network techniques provide efficient representations of quantum many-body states rooted in the entanglement within the wave function. The amount and arrangement of entanglement are mathematically influenced by the chosen network structure and the number of parameters in the tensors.

We can represent the contractions between a set of tensors as a sort of graph:

$$\begin{aligned}
\text{(i)} \sum_i A_i B_i &= \text{---} \textcircled{A} \text{---}^i \text{---} \textcircled{B} \text{---} \\
\text{(ii)} \sum_j A_{i,j} B_{j,k} &= \text{---}^i \text{---} \textcircled{A} \text{---}^j \text{---} \textcircled{B} \text{---}^k \text{---} \\
\text{(iii)} \sum_j A_{i,j} B_j &= \text{---}^i \text{---} \textcircled{A} \text{---}^j \text{---} \textcircled{B} \text{---} \\
\text{(iv)} \sum_{i,j,k,l,n} A_{i,j} B_{i,j,k,l} C_{k,m} D_{l,m,n} &= \text{---} \textcircled{A} \text{---}^i \text{---} \textcircled{B} \text{---}^k \text{---} \textcircled{C} \text{---}^m \text{---} \\
&\quad \text{---}^j \text{---} \textcircled{B} \text{---}^l \text{---} \textcircled{D} \text{---}^n \text{---}
\end{aligned}$$

Figure 2: Some examples of (i) a scalar product, (ii) a matrix-matrix multiplication, (iii) a matrix-vector multiplication and (iv) a more complicated contraction yielding a vector.

3.3 Why we use TNs?

Tensor networks serve as crucial frameworks for understanding the quantum many-body states observed in nature. Unlike traditional methods reliant on wave function coefficients in a given basis, the tensor network approach encapsulates quantum states through interconnected tensor structures, capturing the system's entanglement dynamics effectively. Departing from complex equations, tensor network diagrams provide an intuitive visual representation, offering a more accessible means of grappling with quantum states. This tensor-based language emerges as a natural and powerful tool for deciphering the complexities inherent in the quantum state of matter.

The primary motivation behind this approach stems from the immense size of the Hilbert space. A Hilbert space, being a vector space endowed with an inner product that defines a distance metric, constitutes a complete metric space. However, the sheer vastness of this space poses a formidable challenge. Tensor network states offer a solution by precisely targeting the most pertinent segment of the state: a minuscule, exponentially small fraction within the vast expanse of the Hilbert space. By focusing on this subset, tensor network families navigate the complexity of quantum states efficiently enable more manageable and insightful analyses.

The tensor network approach boasts remarkable flexibility, allowing for the study of diverse systems across various dimensions, ranging from finite to infinite sizes, and exhibiting a plethora of boundary conditions and symmetries. Moreover, it accommodates a broad spectrum of physical phenomena, including bosonic, fermionic, and frustrated spin systems.

3.4 Two families of useful TNs states

Tensor network states offer a foundational concept by representing quantum states as contractions of lower-order tensors. Within this framework, two notable families emerge Matrix

Product States (MPS) and Projected Entangled Pair States (PEPS). Both MPS and PEPS stand out for their versatility, serving as effective numerical approximations for quantum many-body wave functions. They excel in scenarios like capturing low-energy excitations of ground states and modeling local Hamiltonians with precision.

3.4.1 Matrix Product States

Every matrix product state adheres to an area law regarding entanglement entropy. Matrix Product States represent a type of Tensor Network state arranged in a linear array of tensors. In an MPS, the coefficients are expressed as a TN formed by the contraction of N tensors. Visually, an MPS is depicted by a one-dimensional graph featuring N open bonds.

A practical perspective on MPS involves the valence bond construction. Start with pure quantum states $|\psi\rangle \in \mathbb{C}^{\otimes d^N}$ representing a system of N sites, each associated with a d -dimensional Hilbert space. Then, arrange the N entities ('spins') on a ring and assign two virtual spins of dimension D to each. Assume that initially, every pair of neighbouring virtual spins corresponding to different sites are in a maximally entangled state $|I\rangle = \sum_{\alpha=1}^D |\alpha, \alpha\rangle$, often termed as an entangled bond. Then apply a map

$$\mathcal{A} = \sum_{i=1}^d \sum_{\alpha, \beta=1}^D A_{i, \alpha, \beta} |i\rangle \langle \alpha, \beta| \quad (8)$$

to each of the N sites. By writing A_i for the $D \times D$ matrix with elements $A_{i, \alpha, \beta}$ we get that the coefficients of the final state when expressed in terms of a product basis are given by a matrix product $\text{tr}[A_{i_1} A_{i_2} \cdots A_{i_N}]$. In general the dimension of the entangled state $|I\rangle$ and the map $\mathcal{A}$ can both be site-dependent and we write $A_i^{[k]}$ for the $D_k \times D_{k+1}$ matrix corresponding to site $k \in 1, \dots, N$. States obtained in this way have then the form

$$|\psi\rangle = \sum_{i_1, \dots, i_N=1}^d \text{tr} \left[A_{i_1}^{[1]} A_{i_2}^{[2]} \cdots A_{i_N}^{[N]} \right] |i_1, i_2, \dots, i_N\rangle \quad (9)$$

and are called Matrix Product States. Here, equation (9) is a representation of states rather than the characterization of a specific class.

However, typically states are referred to as MPS if they have a MPS-representation with small $D = \max_k D_k$ which does in particular not grow with N . Note that ψ in equation (9) is in general but not normalized and its MPS representation is not unique. Normalization as well as other expectation values of product operators can be obtained from

$$\langle \psi | \bigotimes_{k=1}^N S_k | \psi \rangle = \text{tr} \left[\prod_{k=1}^N E_{S_k}^{[k]} \right] \quad (10)$$

with $E_S^{[k]} \equiv \sum_{i,j=1}^d \langle i | S | j \rangle \bar{A}_i^{[k]} \otimes A_j^{[k]}$

It can be drawn as follow:

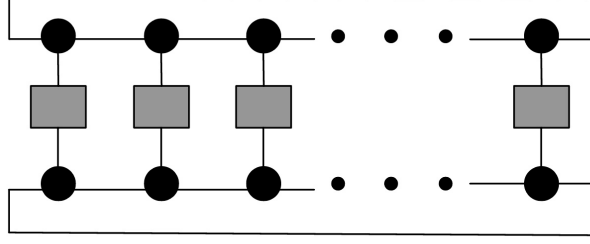


Figure 3: Evaluating an expectation value of an MPS involves contracting the tensor illustrated, where bonds indicate indices being contracted. The matrices linked to each spin are depicted by the circles, with the vertical bond of each matrix representing its physical index, while observables are denoted by squares. It's evident that this contraction can be carried out efficiently.

Here are some basic properties of MPS:

(1) **1d Translational invariance and the thermodynamic limit.** In principle, all tensors in a finite-size Matrix Product State (MPS) could vary, which indicate that the MPS lacks translational invariance. Nevertheless, it is conceivable to enforce translational invariance and extend the MPS to the thermodynamic limit by selecting a fundamental unit cell of tensors that repeats infinitely across the 1D lattice.

(2) **One-dimensional area-law.** This means that the entanglement entropy of a block of sites is bounded by a constant. More precisely, $S(L) = -\text{tr}(\rho_L \log \rho_L) = O(\log D)$ where D is the bond dimensions, with ρ_L the reduced density matrix of the block. This is exactly the behaviour that is usually observed in ground states of gapped 1d local Hamiltonians for large size L of the block: $S(L) \sim \text{constant}$ for $L \gg 1$.

(3) **MPS are dense.** MPS can represent any quantum state within the many-body Hilbert space by sufficiently increasing the value of D . To encompass all states in the Hilbert space, D must be exponentially large relative to the system size. However, it's established that low-energy states of gapped local Hamiltonians in 1D can be efficiently approximated with nearly arbitrary precision using an MPS with a finite value of D . For 1D critical systems, D tends to increase polynomially with the system size. These findings clarify the effectiveness of certain MPS-based techniques for 1D systems, such as the density matrix renormalization group (DMRG).

(4) **Canonical form and the Schmidt decomposition.** For a quantum state $|\psi\rangle$ represented by an MPS with open boundary conditions, there exists a selection of tensors known as the canonical form of the MPS. This is characterized as follows: given an MPS with open boundary conditions (applicable to either finite or infinite systems), it is said to be in its canonical form if for each bond index α , the index aligns with the labeling of Schmidt vectors in the Schmidt decomposition of $|\psi\rangle$ across that index. This can be expressed as:

$$|\psi\rangle = \sum_{\alpha=1}^D \lambda_{\alpha} |\Phi_{\alpha}^L\rangle \otimes |\Phi_{\alpha}^R\rangle \quad (11)$$

In this equation, λ_{α} are Schmidt coefficients ordered into decreasing order ($\lambda_1 \geq \lambda_2 \geq \dots \geq 0$), and the Schmidt vector form orthonormal sets, that is $\langle \Phi_{\alpha}^L | \Phi_{\alpha'}^L \rangle = \langle \Phi_{\alpha}^R | \Phi_{\alpha'}^R \rangle = \delta_{\alpha\alpha'}$.

3.4.2 Projected Entangled Pair State

The family of PEPS is just the natural generalization of MPS to higher spatial dimensions. PEPS are Tensor Network states that correspond to a 2d array of tensors.

A translational invariant PEPS is characterized by a five-index tensor $A \equiv A_{\alpha\beta\gamma\delta}^i$, with a physical index i and virtual or entanglement indices $\alpha, \dots, \delta$. This tensor generates a family of states on systems of arbitrary size $N_h \times N_v$ (and thus asymptotically on the infinite plane) as follows: we arrange the tensor A in an $N_h \times N_v$ grid and contract adjacent indices, as indicated by connected lines in Figure 4 (b), terminating the virtual indices at the boundary with specified boundary conditions. The resulting object now solely depends on the physical indices $\mathbf{i} = (i_{1,1}, \dots, i_{N_h, N_v})$, denoted by $\mathcal{C}_A(\mathbf{i})$. It defines a quantum state, the PEPS, through $|\Psi_{N_h, N_v}\rangle = \sum_{\mathbf{i}} \mathcal{C}_A(\mathbf{i}) |\mathbf{i}\rangle$.

It can be drawn as follow:

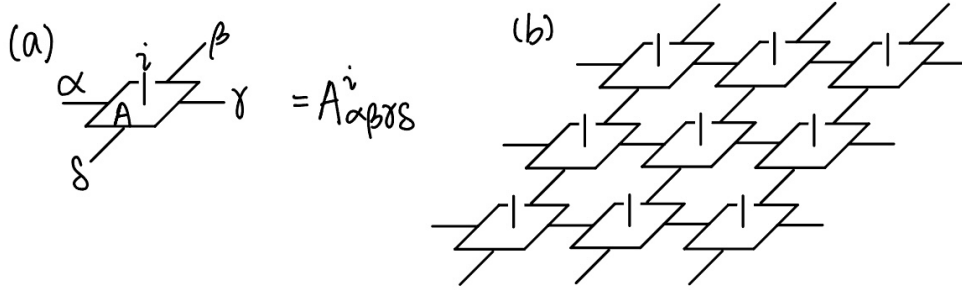


Figure 4: PEPS construction: A five-index tensor (a) is used to describe a wave function on an infinite lattice by contracting the virtual indices $\alpha, \beta, \dots$, as indicated.

Here are some basic properties of PEPS:

(1) **2d Translational invariance and the thermodynamic limit.** Similar to MPS, PEPS is also flexible to choose different tensors, potentially yielding a PEPS devoid of translational invariance. Nonetheless, it remains feasible to enforce translational invariance and extend the PEPS to the thermodynamic limit by adopting a fundamental unit cell repeated throughout the infinite 2D lattice. As anticipated for higher-dimensional systems, translational invariance must be applied across all spatial directions of the lattice.

(2) **2d area-law.** PEPS also obey the area-law scaling of entanglement entropy. While this is primarily observed numerically, there are theoretical justifications supporting this characteristic. For instance, it's widely recognized that $D = 2$ is adequate for managing polynomially decaying correlation functions, and PEPS can accurately approximate thermal states of 2D local Hamiltonians to arbitrary precision.

(3) **PEPS are dense.** Similar to MPS, PEPS constitutes a rich family of states and is capable of representing any quantum state within the many-body Hilbert space by scaling up the bond dimension D . Like MPS, the bond dimension D of a PEPS must grow exponentially with the system's size to span the entire Hilbert space. However, for low-energy states of intriguing 2D quantum models, D is anticipated to remain reasonably small and finite. This observation is not only empirically supported but also underpinned by theoretical arguments.

(4) **No exact canonical form.** In contrast to MPS with open boundary conditions, a PEPS lacks a canonical form in the sense that it's impossible to select an orthonormal basis for all bond indices simultaneously. This limitation arises when dealing with MPS featuring periodic boundary conditions or any scenario involving a loop in the tensor network. Loosely speaking, the presence of a loop in the tensor network disputes the formal division of the network into distinct left and right segments by severing a single index merely. Consequently, attempting to establish a Schmidt decomposition that doesn't formally separate the network into left and right segments becomes nonsensical. In practical terms, this implies that we cannot define orthonormal bases (i.e., Schmidt bases) to the left and right of a given index, thereby eliminating the definition of a canonical form in this context.

3.5 Determining the tensors: finding ground states

MPS and PEPS inherently embody the area law for entanglement entropy within their structure, thereby exhibiting the expected entanglement behaviour for numerous ground states originating from gapped local systems. Given this intrinsic property, it's hardly surprising that MPS and PEPS emerged as indispensable tools for numerical computations. An important consideration arises regarding how to determine the coefficients of tensors within MPS and PEPS. The fundamental approach underlying both families of methods for discovering ground states with tensor networks called variational optimization and imaginary time evolution.

3.5.1 Variational optimization

The variational principle stands as a cornerstone in Quantum Mechanics, aiding in the approximation of the ground state energy of a system and certain excited states. According to the variational principle, the ground-state energy E_0 is never greater than the expectation value of $\mathbf{H}$ computed using the trial wavefunction: that is,

$$E_0 \leq \langle \psi | \mathbf{H} | \psi \rangle \quad (12)$$

Therefore, by adjusting ψ until the expectation value of $\mathbf{H}$ is minimized, we can derive approximations for both the wavefunction and the ground state energy.

In a Hilbert space with a Hermitian operator $\mathbf{H}$ known as the Hamiltonian, we examine the discrete spectrum of $\mathbf{H}$ and a basis composed of $|Y_i\rangle$. Subsequently, the characteristic equation for $\mathbf{H}$ would establish the relationship:

$$\mathbf{H}|Y_i\rangle = E_i|Y_i\rangle \quad (13)$$

where E_i is the i -th eigenvalue (physically the i -th energy level), i ranges within the N dimensional space of states and $|Y_i\rangle$ is the corresponding eigenstate.

In order to find the ground state, we need to minimize the energy expectation value

$$E_\alpha = \frac{\langle \psi_\alpha | H | \psi_\alpha \rangle}{\langle \psi_\alpha | \psi_\alpha \rangle} \quad (14)$$

Where H represents the model Hamiltonian under consideration, $\alpha = \{\alpha_1, \alpha_2, \dots, \alpha_p\}$ denotes a set of p variational parameters present in the wave function $\psi_\alpha(x)$, with $|x\rangle$ denoting a configuration with specified electron spins and positions. The symbol $\langle \rangle$ signifies the quantum expectation value over the state ψ_α , such that $E_\alpha = \langle H \rangle$. In the Newton method, to refine the variational parameters, $E_{\alpha+\gamma}$ is quadratically expanded in terms of γ and minimized. Subsequently, the parameters are updated to $\alpha' = \alpha + \gamma$, and the iteration continues until γ becomes negligible.

An iterative method employs a strategy akin to the one described, exceeds the standard Newton method significantly. In each iteration and for every variational parameter α_h , we introduce an operator O_k that is diagonal in the configuration basis, with diagonal elements given by:

$$O_k(x) = \frac{\delta_{\alpha_k} \psi_\alpha(x)}{\psi_\alpha(x)} \quad (15)$$

In order to simplify the derivation we assume that these operators O_k appear in the wave function in the simple exponential form $\exp[\sum_k \alpha_k (O_k - \langle O_k \rangle)]$, where the operators O_k are density-density or spin-spin correlations. We have subtracted from each O_k the corresponding average value $\langle O_k \rangle$, which provides an irrelevant multiplicative constant in the wavefunction. For the variational parameters corresponding to the uncorrelated part $|D\rangle$ (mean-field wavefunction), this exponential form is not exactly fulfilled, but is valid only for small changes of the variational parameters $|\psi_{\alpha+\gamma}\rangle \propto \exp[\sum_k \gamma_k (O_k - \langle O_k \rangle)] |\psi_\alpha\rangle$, namely within linear order in γ .

We can write $|\psi_{\alpha+\gamma}\rangle$ in a more general form that depends on a further parameter β

$$|\psi_{\alpha+\gamma}\rangle \simeq \left[1 + \sum_k \gamma_k (O_k - \langle O_k \rangle) + \frac{\beta}{2} \sum_{k,k'} \gamma_k \gamma_{k'} (O_k - \langle O_k \rangle) (O_{k'} - \langle O_{k'} \rangle) \right] |\psi_\alpha\rangle \quad (16)$$

This expansion is true only up to linear (quadratic) order in γ for $\beta \neq 1$. The parameter β can be leveraged to enhance the efficiency of the minimization approach within the variational Monte Carlo method. As the minimum is approached, the nonlinear contribution proportional to β , along with higher-order terms, diminish and become inconsequential for optimizing the wavefunction.

By substituting expression (16) into equation (14), the energy can be systematically expanded in powers of γ_k

$$\Delta E = - \sum_k \gamma_k f_k + \frac{1}{2} \sum_{k,k'} \gamma_k \gamma_{k'} [S_h + (1 + \beta)G]^{k,k'} \quad (17)$$

where

$$S_h^{k,k'} = \langle [O_k, [H, O_{k'}]] \rangle \quad (18)$$

$$G^{k,k'} = 2 \langle (H - E_\alpha) (O_k - \langle O_k \rangle) (O_{k'} - \langle O_{k'} \rangle) \rangle \quad (19)$$

$$f_k = -\partial_{\alpha_k} E_\alpha = -2 \langle (H - E_\alpha) O_k \rangle \quad (20)$$

In these equations, we utilize the Hermitian nature of all involved operators, implying, for instance, that $\langle O_k H \rangle = \langle H O_k \rangle$. f_k denotes the forces acting on the variational parameters, which vanish under the condition of minimum energy. $S_h^{k,k'}$ represents the excitation matrix elements corresponding to the operators O_k , while $G^{k,k'}$ incorporates additional contributions arising when the wavefunction is not exact ($\langle H - E_\alpha \rangle \neq 0$). The square brackets denote the commutator. Subsequently, the wavefunction parameters are iteratively adjusted as $\alpha_k \rightarrow \alpha_k + \gamma_k$, by minimizing equation (17) whenever $S_h + (1 + \beta)G$ is positive definite. The minimum energy is attained for

$$\gamma = B^{-1} \mathbf{f} \quad (21)$$

with $B = S_h + (1 + \beta)G$.

3.5.2 Imaginary time evolution

Imaginary time evolution method holds significant mathematical power. It bridges Euclidean space, quantum mechanics, statistical mechanics, and static problems to dynamic ones by substituting real time with imaginary time. A crucial aspect of imaginary-time schemes involves determining the optimal truncations of enlarged bonds which is similar to contraction schemes in 3D tensor networks (TNS), . For a system evolving under Hamiltonian H in real time t , the propagator is e^{-iHt} . In contrast, in imaginary time, denoted as $\tau = it$, the corresponding propagator becomes $e^{-H\tau}$ representing a non-unitary operator.

We focus on many-body systems that are described by Hamiltonians $H = \sum_i \lambda_i h_i$, with real coefficients λ_i , and observables h_i that are tensor products of Pauli matrices. We assume that the number of terms in this Hamiltonian scales polynomially with the system size. Given an initial state $|\psi\rangle$, the normalized imaginary time evolution is defined by

$$|\psi(\tau)\rangle = A(\tau)e^{-H\tau}|\psi(0)\rangle, \quad (22)$$

where $A(\tau) = \frac{1}{\sqrt{\langle\psi(0)|e^{-2H\tau}|\psi(0)\rangle}}$ is a normalization factor.

Next, we use the imaginary time evolution technique to address the task of determining the ground-state energy of a many-body Hamiltonian H . Our goal is to find the values of the parameters $\vec{\theta}$, which is used to minimize the expectation value of Hamiltonian

$$E_{min} = \min_{\vec{\theta}} \langle\psi(\vec{\theta})|H|\psi(\vec{\theta})\rangle, \quad (23)$$

where $|\psi(\vec{\theta})\rangle = V(\vec{\theta})|\bar{0}\rangle$ is our variational trial state.

To proceed, we divide the local Hamiltonians on the square lattice into four groups: $\hat{H}_{e,e} = \sum_{\text{even } i,j} \hat{H}^{[i,j;i,j+1]} + \hat{H}^{[i,j;i+1,j]}$, $\hat{H}_{o,o} = \sum_{\text{odd } i,j} \hat{H}^{[i,j;i,j+1]} + \hat{H}^{[i,j;i+1,j]}$, $\hat{H}_{e,o} = \sum_{\text{even } i, \text{ odd } j} \hat{H}^{[i,j;i,j+1]} + \hat{H}^{[i,j;i+1,j]}$, and $\hat{H}_{o,e} = \sum_{\text{odd } i, \text{ even } j} \hat{H}^{[i,j;i,j+1]} + \hat{H}^{[i,j;i+1,j]}$. One can see that each two terms in one group commute to each other. The evolution operator for an infinitesimal imaginary-time step ($\tau \rightarrow 0$) then can be written as

$$\hat{U} = \exp(-\tau \hat{H}) = \exp(-\tau \hat{H}^{[e,e]}) \exp(-\tau \hat{H}^{[o,o]}) \exp(-\tau \hat{H}^{[e,o]}) \exp(-\tau \hat{H}^{[o,e]}) + O(\tau^2) \quad (24)$$

Assume translational invariance to the Hamiltonian, i.e., $\hat{H}^{[i,j]} = \hat{H}^{[i+1,j]}$. The element of two-body evolution operator is a fourth-order tensor $U_{s_i s_j s'_i s'_j} = \langle s'_i s'_j | \exp(-\tau \hat{H}^{[two]}) | s_i s_j \rangle$. Implement QR decomposition on U as

$$U_{s_i s_j s'_i s'_j} = \sum_{\alpha} L_{s_i s'_i, \alpha} R_{s_j s'_j, \alpha} \quad (25)$$

Then the two tensors $T^{[L]}$ and $T^{[R]}$ that form the imaginary projected entangled pair operator of $\hat{U}$ is obtained as

$$T_{ss', a_1 a_2 a_3 a_4}^{[L]} = \sum_{s_1 s_2 s_3} L_{ss_1, a_1} L_{s_1 s_2, a_2} L_{s_2 s_3, a_3} L_{s_3 s', a_4} \quad (26)$$

$$T_{ss', a_1 a_2 a_3 a_4}^{[R]} = \sum_{s_1 s_2 s_3} R_{ss_1, a_1} R_{s_1 s_2, a_2} R_{s_2 s_3, a_3} R_{s_3 s', a_4} \quad (27)$$

The four L s (or R s) in $T^{[L]}$ (or $T^{[R]}$) correspond to the evolution operators of the two-body terms in $\hat{H}^{[e,e]}$, $\hat{H}^{[o,o]}$, $\hat{H}^{[e,o]}$, $\hat{H}^{[o,e]}$ in equation (23), respectively.

While the tensor network for imaginary time evolution with the imaginary Projected Entangled Pair Operator (iPEPO) forms a cubic TN, one can also employ the tensor U , which results in a 3D but non-cubic TN structure. Without losing generality, we will utilize the iPEPO to illustrate the algorithms for contracting a cubic TN. However, it is worth noting that this algorithm can be readily extended to handle statistical models on cubic lattices or other problems that can be represented as the contraction of a cubic TN.

The evolution $\hat{U}|\psi\rangle$ is to contract the iPEPO (one layer of the tensors) to the iPEPS. In accordance to the translational invariance of the iPEPO, the iPEPS is also formed by two inequivalent tensors (denoted by P^L and P^R). Locally, the tensors in the evolved iPEPS are given as

$$\tilde{P}_{s,\tilde{\alpha}_1,\tilde{\alpha}_2,\tilde{\alpha}_3,\tilde{\alpha}_4}^{[L]} = \sum_{s'} T_{ss',a_1a_2a_3a_4}^{[L]} P_{s',\alpha_1\alpha_2\alpha_3\alpha_4}^{[L]} \quad (28)$$

$$\tilde{P}_{s,\tilde{\alpha}_1,\tilde{\alpha}_2,\tilde{\alpha}_3,\tilde{\alpha}_4}^{[R]} = \sum_{s'} T_{ss',a_1a_2a_3a_4}^{[R]} P_{s',\alpha_1\alpha_2\alpha_3\alpha_4}^{[R]} \quad (29)$$

Using composite indices $\tilde{\alpha}_x = (a_x, \alpha_x)$ where $x = 1, 2, 3, 4$, it's evident that the bond dimensions of the new tensors increase by a factor of $\dim(a_x)$. To manage this growth, a dimension cut-off χ needs to be predetermined. Once the bond dimensions exceed χ , approximations are introduced to reduce them back to χ , ensuring bounded computational cost during the iterative evolution of the imaginary Projected Entangled Pair States (iPEPS). Upon convergence of the iPEPS, the ground state is thought reached. Hence, a crucial step in both imaginary-time schemes and similar contraction schemes of 3D tensor networks is to determine the optimal truncations of the expanded bonds.

4 The restricted Boltzmann machine

Despite the relative success of Tensor Networks (TNs) in dealing with many-body systems, mainly in one dimension, researchers are still looking for practical ways to overcome the limitations of the subconvolution law and to numerically handle highly entangled systems in one, two, or even three dimensions. Restricted Boltzmann Machine (RBM)-based wavefunction representations have recently demonstrated this capability, but the number of parameters they require is incompatible with current optimization methods.

4.1 Definition of RBM

A Restricted Boltzmann Machine (RBM) is a probabilistic model characterized by a layer of hidden variables, enabling it to capture highly expressive marginal distributions. RBMs, widely utilized as unsupervised learning techniques, have found extensive applications across diverse fields including pattern recognition, computer vision, and signal processing. However, traditional RBMs are tailored for vector data and are not inherently equipped to handle matrices and higher-dimensional data, which are prevalent in real-world scenarios. The restricted Boltzmann machine is composed of visible units v_i that make up the data space and extra degrees of freedom called hidden units h_j . The regions where each degree of freedom exists are called visible and hidden layers. Both v_i and h_j take two states, we define the values of the two states as $v_i = \pm 1$, $h_j = \pm 1$.

Named the restricted Boltzmann machine due to the limitation that there is no intra-layer interaction within the visible and hidden layers. Since there is no interaction among hidden

spins, we can analytically trace out hidden spins, thus obtaining the marginal probability distribution

$$\tilde{p}(v) \propto \exp\left(\sum_i a_i v_i\right) \times \prod_j 2\cosh(b_j + \sum_i W_{ij} v_i) \quad (30)$$

Here, a_i and b_j represent the magnetic fields acting on the visible and hidden spins, respectively, while W_{ij} denotes the classical interaction between these spins. In the realm of machine learning, the terms associated with a_i and b_j are commonly referred to as bias terms. From this equation, we observe that the Restricted Boltzmann Machine (RBM) takes the visible spin configuration v as input and generates the marginal probability $\tilde{p}(v)$ as output. This $\tilde{p}(v)$ serves to approximate the probability distributions across the visible spin configurations. In the case of the RBM, any arbitrary probability distribution can be faithfully replicated with arbitrary precision by introducing an exponentially large number of hidden spins ($M \sim 2^N$), which is comparable to the number of visible spin configuration patterns 2^N .

Graphically, RBM can be represented as follow:

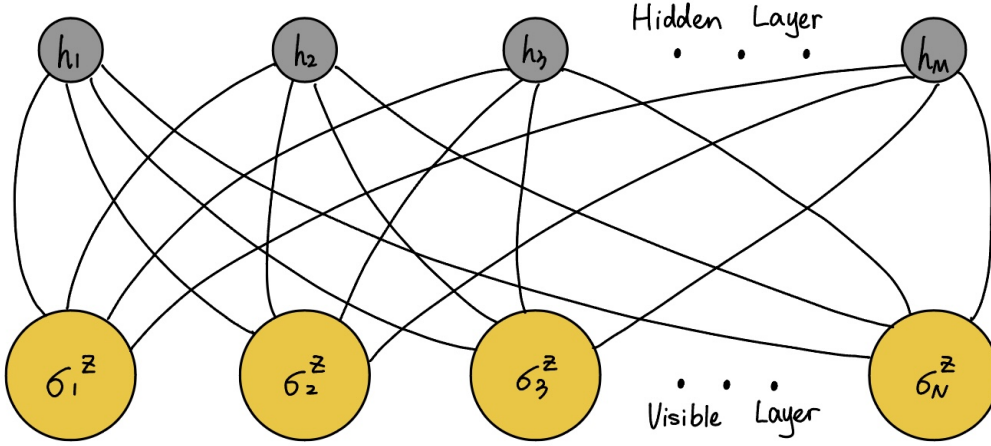


Figure 5: Restricted Boltzmann Machine representation.

The illustration depicts an artificial neural network encoding a many-body quantum state comprising N spins. It shows a restricted Boltzmann machine architecture, comprising a collection of N visible artificial neurons (yellow dots) and a set of M hidden neurons (grey dots). For every configuration of the many-body spin system $S = (\sigma_1^z, \sigma_2^z, \dots, \sigma_N^z)$, the artificial neural network calculates the value of wave function $\Psi(S)$.

4.2 Wave function based on Boltzmann machines

We use RBMs, which are energy-based generative neural networks, as a variational ansatz for quantum many-body spin systems. The RBM models a distribution $\Psi(x_1, \dots, x_N) \equiv \Psi(\mathbf{x})$ of the variables $x_{i \in 1, \dots, N}$, characterized by the energy function

$$E_{RBM}(\mathbf{x}, \mathbf{h}; \mathcal{W}) = \sum_i^N \sum_j^M w_{ij} x_i h_j + \sum_i^N a_i x_i + \sum_j^M b_j h_j \quad (31)$$

Here, $h_{j \in 1, \dots, M} \in -1, 1$ is a set of binary latent variables. The set $\mathcal{W} = w_{ij}, a_i, b_j$ are variational parameters: W_{ij} are the weights connecting variables x_i and h_j , and $a_i(b_j)$ are the biases of the physical (latent) variables $x_i(h_j)$. The ratio of the number of latent variables M and the number of physical variables N , defined by $\alpha \equiv M/N$, is a measure of the complexity of the model.

The RBM is introduced as a variational ansatz for quantum many-body wave functions by modelling the probability amplitudes $\Psi(\mathbf{x})$ of the wave function $|\Psi\rangle = \sum(X)\Psi(\mathbf{x})|\mathbf{x}\rangle$ as the marginalized Boltzmann distribution

$$\Psi(\mathbf{x}; \mathcal{W}) \equiv \sum_{\mathbf{h}} e^{-E_{RBM}(\mathbf{x}, \mathbf{h}; \mathcal{W})}, \mathcal{W} \subset \mathbb{C} \quad (32)$$

This model extends the Restricted Boltzmann Machine (RBM) concept, which typically represents classical probability distributions by marginalizing over hidden variables of a Boltzmann distribution. Here, x_i is not constrained to binary variables but remains discrete. Instead of $w_{ij}x_i$, a set of general discrete functions $w_{ij}[x_i]$ is substituted. This nonlinearly transforms the variables x_i before constructing the RBM energy equation (31).

In order to find eigenstates of a Hamiltonian $\hat{H}$, we use the variational principle to minimize the energy function

$$E(\Psi; \mathcal{W}) = \frac{\langle \Psi | \hat{H} | \Psi \rangle}{\langle \Psi | \Psi \rangle} \quad (33)$$

with respect to the parameters $\mathcal{W}$, for which we use the stochastic reconfiguration method*. This method is closely related to the imaginary-time Hamiltonian evolution of quantum states, and hence the stochastic reconfiguration method greatly stabilizes the optimization.

*The stochastic reconfiguration method makes use of information about the geometry of quantum states (Fubini-Study metric). The Fubini-Study metric is a metric on the space of quantum states, whose form is given by

$$\mathcal{F}[|\psi\rangle, |\phi\rangle] := \arccos \sqrt{\frac{\langle \psi | \phi \rangle \langle \phi | \psi \rangle}{\langle \psi | \psi \rangle \langle \phi | \phi \rangle}} \quad (34)$$

For normalized quantum states $|\bar{\psi}\rangle = \frac{|\psi\rangle}{\sqrt{\langle \psi | \psi \rangle}}$ and $|\bar{\phi}\rangle = \frac{|\phi\rangle}{\sqrt{\langle \phi | \phi \rangle}}$, the Fubini-Study metric reads

$$\mathcal{F}[|\bar{\psi}\rangle, |\bar{\phi}\rangle] = \arccos(|\langle \bar{\psi} | \bar{\phi} \rangle|) \quad (35)$$

This can be understood as the angle between two complex vectors. Thus, the Fubini-Study metric is sometimes referred to the quantum angle. Now examine the infinitesimal form of this metric; for a variational quantum state $|\psi_\theta\rangle$ characterized by the parameter set θ , it is given by

$$\mathcal{F}^2[|\psi_{\theta+\delta\theta}\rangle, |\psi_\theta\rangle] = \sum_{kl} S_{kl}(\theta) \delta\theta_k \delta\theta_l \quad (36)$$

$S_{kl}(\theta) = \text{Re}[Q_{kl}(\theta)]$ is the Fubini-Study metric tensor and is given as the real part of so called quantum geometric tensor $Q_{kl}(\theta)$:

$$Q_{kl}(\theta) = \frac{\langle \partial_k \psi_\theta | \partial_l \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle} - \frac{\langle \partial_k \psi_\theta | \psi_\theta \rangle \langle \psi_\theta | \partial_l \psi_\theta \rangle}{\langle \psi_\theta | \psi_\theta \rangle^2} \quad (37)$$

The Fubini-Study metric can be regarded as a quantum version of the Fisher information metric and thus has a close relationship with information geometry.

As the stochastic reconfiguration optimization is closely related to the imaginary-time evolution of quantum states, the update rule can be derived by trying to reproduce the infinitesimal imaginary-time evolution $e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle$ by the variational state with updated parameters $|\psi_{\theta+\delta\theta}\rangle$:

$$\delta\tilde{\theta} = \arg \min_{\delta\theta} \mathcal{F}[e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle, |\psi_{\theta+\delta\theta}\rangle] \quad (38)$$

This equation is equivalent to:

$$\delta\tilde{\theta} = \arg \max_{\delta\theta} F^2[e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle, |\psi_{\theta+\delta\theta}\rangle] \quad (39)$$

Here, $F^2[e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle, |\psi_{\theta+\delta\theta}\rangle]$ is the fidelity between $e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle$ and $|\psi_{\theta+\delta\theta}\rangle$:

$$F^2[e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle, |\psi_{\theta+\delta\theta}\rangle] = \frac{\langle\psi_\theta|e^{-2\delta\tau\mathcal{H}}|\psi_{\theta+\delta\theta}\rangle\langle\psi_{\theta+\delta\theta}|e^{-2\delta\tau\mathcal{H}}|\psi_\theta\rangle}{\langle\psi_\theta|e^{-4\delta\tau\mathcal{H}}|\psi_\theta\rangle\langle\psi_{\theta+\delta\theta}|\psi_{\theta+\delta\theta}\rangle} \quad (40)$$

Expanding this equation to the second order for minute quantities yields

$$F^2 = 1 - \left(\sum_{k,l} \delta\theta_k S_{kl} \delta\theta_l + 2\delta\tau \sum_k g_k \delta\theta_k + 4\delta\tau^2 E_{var} \right) \quad (41)$$

Here, S_{kl} is the Fubini-Study metric tensor. g_k is the gradient of energy with respect to the variational parameter, whose detailed form reads

$$g_k = 2Re \left(\frac{\langle\psi_\theta|\mathcal{H}|\partial_k\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} - \frac{\langle\psi_\theta|\mathcal{H}|\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} \frac{\langle\psi_\theta|\partial_k\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} \right) \quad (42)$$

E_{var} is the variance of the total energy

$$E_{var} = \frac{\langle\psi_\theta|\mathcal{H}^2|\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} - \left(\frac{\langle\psi_\theta|\mathcal{H}|\psi_\theta\rangle}{\langle\psi_\theta|\psi_\theta\rangle} \right)^2 = \langle\mathcal{H}^2\rangle - \langle\mathcal{H}\rangle^2 \quad (43)$$

By considering the stationary condition in equation (41), we obtain the following equation:

$$\delta\theta_k = -\delta\tau \sum_l S_{kl}^{-1} g_l \quad (44)$$

Hence, the update rule shows following:

$$\delta\theta_k^{(t)} = -\delta\tau \sum_l (S_{kl}^{(t)})^{-1} g_l^{(t)} \quad (45)$$

where t indicates the iteration number of optimization. $g_l^{(t)}$ is the gradient of the total energy (loss function) with respect to the variational parameters

$$g_k^{(t)} = \left. \frac{\partial E_\theta}{\partial \theta_k} \right|_{\theta=\theta^{(t)}} \quad (46)$$

The move in equation (45) can also be interpreted as the steepest descent direction with respect to the quantum information geometry.

5 Comparison between restricted Boltzmann machine and tensor networks

5.1 Representability and entanglement entropy

The neural network method complements the tensor network approach, which includes techniques like the Density Matrix Renormalization Group (DMRG) utilizing Matrix Product States (MPS). This synergy arises from the fact that tensor networks, often referred to as universal approximators in the context of large bond dimensions, encompass a wide array of functions and patterns. It aligns with the flexibility and adaptability inherent in neural networks.

For tensor networks with finite bond dimensions, their representability is characterized by entanglement entropy effectively. Tensor networks adhering to finite bond dimensions conform to the area law of entanglement entropy. Specifically, the maximum entanglement entropy achievable by an MPS with a bond dimension of χ is $\log \chi$. Consequently, tensor networks excel at approximating quantum states characterized by area-law entanglement entropy.

However, when it comes to Boltzmann machine wave functions, and more generally, neural network wave functions with a finite number of hidden degrees of freedom, the absence of robust metrics presents a significant hurdle. Consequently, there is a lack of a comprehensive theory elucidating how the accuracy of approximation improves as hidden degrees of freedom increase, and how it asymptotically approaches the limit of universal approximation.

5.2 Transformation between MPS and Boltzmann machines

The exploration of transforming neural network wave functions into tensor network wave functions, or vice versa offers an intriguing avenue to better understand the interplay between these two paradigms. To illustrate, we delve into the transformation rule between Matrix Product States (MPSs) and Boltzmann machine quantum states.

5.2.1 From RBM to MPS

Indeed, a conversion law exists from Restricted Boltzmann Machine (RBM) representations to Matrix Product States (MPS). As discussed in section 5.1, quantum states represented by RBMs can manifest volume-law entanglement entropy. Given that the maximum entanglement entropy of an MPS with a bond dimension χ is $\log \chi$, representing quantum states with volume-law entanglement entropy necessitates an exponentially large bond dimension.

5.2.2 From MPS to RBM

Concerning the inverse transformation, there is not a universal rule for converting from Matrix Product States (MPS) to Restricted Boltzmann Machines (RBM). Nonetheless, due to its universal approximation capability, the RBM can emulate any given MPS as the number of hidden spins grows large.

6 Challenges and Outlook

In recent years, the integration of machine learning with quantum many-body problems has seen significant advancement. Its utilization spans various domains, encompassing numerical simulation and data analysis. Machine learning techniques have showed their potential by

outperforming existing methods in specific problem domains. However, the full extent of their prowess in this domain remains partially realized, with several unresolved challenges awaiting solutions.

Numerous avenues for future research emerge on the horizon. For instance, recent strides in machine learning, such as deep network architectures and convolutional neural networks, hold promise as foundational elements for more sophisticated neural network quantum states, potentially enhancing their expressive capabilities. Moreover, extending our methods to encompass quantum systems beyond interacting spins appears to be relatively straightforward. In this regard, the potential applications to address complex inquiries concerning interacting fermions in two dimensions are already foreseeable. In contrast to tensor network states, neural-network quantum states inherently embody nonlocal correlations, potentially leading to significantly more concise representations of many-body quantum states. A systematic examination of the entanglement properties of Neural Network Quantum States (NQS) could thus introduce novel concepts in quantum information theory.

While machine learning and AI algorithms rooted in quantum physics hold immense potential, they are currently in the nascent stages of research, accompanied by various challenges and constraints. Primarily, the hardware implementations of quantum computing are grappling with technical hurdles, necessitating the development of more stable and dependable quantum bits. Moreover, the encoding and decoding procedures of quantum algorithms demand a profound level of expertise, thus raising the bar for algorithm researchers. Furthermore, the practical disposition of quantum algorithms hinges upon substantial experimental validation and extensive support with large-scale data.

In summary, machine learning and algorithms based on quantum physics have broad application prospects and far-reaching impacts. With the continuous development of quantum computing technology and in-depth research on algorithms, it is believed that quantum physics-based machine learning and algorithms will play an important role in various fields, promoting scientific and technological progress and social development. We expect to see more practical applications of quantum physics-based machine learning and algorithms in the future, bringing more convenience and innovation to human life.

7 Acknowledgements

I extend my deepest gratitude to my supervisor, Dr. Itamar Yaakov, for his invaluable insights and enlightening discussions throughout this project. Additionally, I wish to express my sincere appreciation to my personal academic tutors, Prof. Ruben Sanchez-Garcia and Prof. Jan Spakula, for their guidance and support. Lastly, I am profoundly grateful to the University of Southampton for providing me with the resources and environment conducive to my academic growth and personal development. It is an honour to be a student here, and I take great pride in being a part of this institution. Of course, I must also express my gratitude to myself for my perseverance and dedication throughout this journey, as well as to my family and friends for their unwavering support.

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